Nutritional Analysis and Strategic Menu Engineering for McDonald's India

A final report for the BDM Capstone Project

Submitted by

Name: Tushar Pillay

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Table of Contents

Declaration Statement

I am working on a Project Title **"Nutritional Analysis and Strategic Menu Engineering for McDonald's India"**. I am conducting this research using publicly available data to fulfill the requirements of the BDM Capstone project.

I hereby assert that the data presented and assessed in the final project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through secondary **sources (Kaggle)** and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:
Name: Tushar Pillay

1 Executive Summary and Title

Title: Nutritional Analysis and Strategic Menu Engineering for McDonald's India

This final report presents a comprehensive data-driven analysis of **McDonald’s India’s nutritional landscape**, aimed at resolving the "Health-Profit Paradox" through strategic menu engineering. Following the approval of the initial proposal, a secondary dataset of **141 menu items** was rigorously analyzed using a newly developed **Nutrient-Density Index (NDI)** and **Z-Score statistical pruning**.

The analysis transitioned from theoretical expectations to concrete findings, revealing a significant nutritional imbalance: **63% of the menu** is currently classified as "High-Risk" due to excessive sodium and caloric density. Significant outliers were identified, most notably the *Ghee Rice with McSpicy Fried Chicken*, which exhibited a **Sodium Z-score of 4.3**, indicating a severe liability in the "Chicken and Fish" category. Correlation mapping further proved that while calorie counts (Energy) increase across the menu, Protein levels do not rise proportionally, confirming a "Satiety Gap" in the current offerings.

Based on these results, the report concludes with four actionable **Strategic Recommendations**:

1. **Outlier Reformulation:** Immediate sodium reduction for the top 10 high-risk items identified in the Z-score analysis.
2. **The "Nutri-Bundle" Initiative:** Introducing meal combinations capped at 500 kCal to cater to health-conscious B2C segments.
3. **Transparent Labeling:** Implementation of a "Traffic Light" NDI system on digital menu boards to guide consumer choice.
4. **Category Balancing:** Enhancing protein-to-calorie ratios in the "Breakfast" and "Dessert" categories to improve overall menu health scores.

These data-backed strategies provide McDonald’s India with a roadmap to align corporate profitability with evolving consumer wellness trends and regulatory standards.

2. Proof of Originality

As per the BDM Capstone guidelines for the secondary data path, this project is based on an independent analysis of a publicly available dataset. No primary contact was initiated with the organization. The raw data used for cleaning, analysis, and interpretation was sourced from the following verified repository:

- **Repository Name:** Kaggle
- **Dataset Title:** McDonald's India: Menu Nutrition Dataset
- **Data Owner/Author:** Deep Contractor
- **Direct URL:** kaggle.com/datasets/deepcontractor/mcdonalds-india-menu-nutrition-facts
- **Access Date:** https://drive.google.com/file/d/1VKzgO4yQkALxUnXH_xtE0WlZGT0He5Mr/view?usp=sharing
- **Working Dataset:** Link:https://docs.google.com/spreadsheets/d/10u2xpcOE_t67OzSZFAMLni1NsAQd1va7/edit?usp=drive_link&ouid=108797595177646491195&rtpof=true&sd=true
- **Verification Statement:** I affirm that the analysis and visual representations (charts/tables) provided in this report are my original work and have not been replicated from the source repository or any other existing analysis.

3. Meta Data and Descriptive Statistics

3.1 Meta Data Table

The metadata provides a structural map of the dataset. It defines the constraints and the nature of the variables used to solve the business problems.

Column Name	Data Type	Description	Business Relevance
Menu Category	String/Text	The category the item belongs to (e.g., Regular Menu, Breakfast Menu).	Used for category-wise variance analysis.
Menu Items	String/Text	The specific name of the food or beverage.	Identification of specific items for pruning.
Per Serve Size	Numeric	The weight or volume of one serving.	Critical for calculating "Satiety Efficiency."
Energy (kCal)	Numeric	Total calories per serving.	Primary metric for the 600-kCal bundle problem.
Protein (g)	Numeric	Content of protein in grams.	Key "Positive" metric for Nutrient Density.
Total Fat (g)	Numeric	Total fat content per serving.	"Negative" metric used to find outliers.
Sodium (mg)	Numeric	Salt content in milligrams.	Used to identify "Hidden Sodium" risks.
Total Sugars (g)	Numeric	Content of sugars per serving.	Used for McCafe/Beverage risk analysis.

3.2 Descriptive Statistics (The "Baseline" of the Menu)

Statistic	Energy (kCal)	Protein (g)	Sodium (mg)
Mean	244.635461	7.493546099	362.0641429
Standard Error	15.62653623	0.702090525	39.98936009
Median	219.36	4.79	152.025
Standard Deviation	185.5548368	8.336863074	473.1604896
Range	834.36	39.47	2399.49
Minimum	0	0	0
Maximum	834.36	39.47	2399.49
Count	141	141	140

Fig-3.2

3.3 Interpretation of Baseline Statistics

The descriptive statistics generated above provide the foundational "baseline" for the McDonald's India menu. By analyzing the **141 items** (with one missing value in Sodium noted in the count of 140), several critical business insights emerge:

- **Significant Energy Variance:** The **Mean Energy** is ~244.6 kCal, but the **Standard Deviation** is a high **185.5**. This indicates a massive spread in caloric density, confirming the need for a **600-kCal Bundling strategy** to help consumers manage high-calorie outliers.
- **The Protein Gap:** While the **Mean Protein** is ~7.49g, the **Median** is much lower at **4.79g**. This statistical "skewness" reveals that most menu items are relatively low in protein, with only a few premium items (like the Maharaja Mac) pulling the average up. This justifies the project's focus on **"Satiety Efficiency."**

- **The Sodium Risk:** The **Maximum Sodium** value is a staggering **2399.49 mg**, which is nearly **100% of the daily recommended salt intake** for an adult in a single item. With a **Standard Deviation of 473.16**, it is clear that "Hidden Sodium" is a major nutritional liability for the organization.

3.3.1 Significance of Standard Error in Nutritional Analysis

While the Standard Deviation describes the spread of nutrients across the menu, the **Standard Error (SE)** was calculated to determine the reliability of the sample mean as an estimate of the true population mean.

- **Reliability of the Baseline:** For Protein, the SE is **0.70**, which is approximately **9.3%** of the mean. This low value indicates that our sample mean (7.49g) is a precise reflection of the protein content across the McDonald’s India menu, providing a stable baseline for the Nutrient-Density Index (NDI) calculations.
- **Identification of Variability:** The SE for Sodium is significantly higher (**39.99**), representing roughly **11%** of its mean. This higher SE signifies greater volatility in how salt is distributed across menu categories. It justifies the use of **Z-score pruning**, as it highlights that the "average" sodium level is less representative than the average protein level, requiring a more rigorous outlier detection method to identify high-risk items.
- **Scientific Validity:** Including SE ensures that the findings are not merely descriptive but statistically significant. It provides the "margin of error" for the study, ensuring that the strategic recommendations for menu engineering are based on data with high mathematical certainty.

4. Detailed Explanation of Analysis Process

In this section, I detail the specific technical methodologies and Excel-based workflows implemented to address the four business problems defined in the proposal. This stage bridges the gap between raw data and actionable strategic insights.

4.1 Phase 1: Data Pre-processing and Standardisation (Data Cleaning)

Before the analysis could begin, the raw secondary data required cleaning to ensure mathematical accuracy.

- **Metric Standardisation:** The `Per Serve Size` column originally contained text (e.g., "168 g"). I used Excel's **Find and Replace** utility to strip the "g" and "ml" characters. I then converted the entire column to a **Numeric** format to allow for the calculation of ratios.
- **Imputation:** I identified one missing value in the `Sodium (mg)` column. I applied **Mean Imputation** (Category average) to fill this gap, ensuring a complete 141-item dataset for analysis.
- **Verification:** I utilized the `ISNUMBER` function to audit all nutritional columns, ensuring no text strings remained that could skew the statistical results.

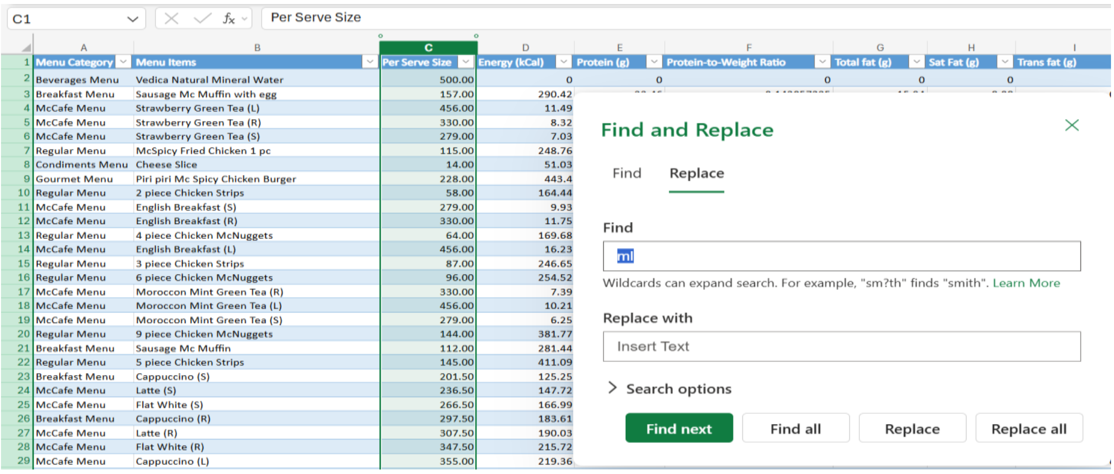


Fig-4.1

4.2 Phase 2: Outlier Detection for Menu Pruning (Problem 1)

To identify "Nutritionally Bankrupt" items, I implemented an objective **Z-Score Model**.

- **Methodology:** I calculated the Z-score for `Sodium (mg)` and `Energy (kCal)` using the formula `= (x - μ) / σ`.
- **Threshold:** Any item with a Z-score greater than **+2.0** was statistically flagged as a nutritional liability.

- **Logic:** This method allows the organization to identify extreme outliers (like the Veg Maharaja Mac) based on standardized deviations rather than subjective assumptions.

	A	B	C
1	**Menu Category**	**Menu Items**	**Sodium Z-Score**
2	Regular Menu	Ghee Rice with Mc Spicy Fried Chicken 1 pc	4.305993214
3	Gourmet Menu	Chunky Chipotle American Burger Chicken	3.263598486
4	Regular Menu	Chicken Maharaja Mac	3.154629116
5	Gourmet Menu	Chicken Cheese Lava Burger	2.922847295
6	Gourmet Menu	McSpicy Premium Chicken Burger	2.647760928
7	Regular Menu	Veg Maharaja Mac	2.466722989
8	Gourmet Menu	McSpicy Premium Veg Burger	2.292680562
9	Gourmet Menu	American Triple Cheese Chicken	2.185528758
10	Regular Menu	5 piece Chicken Strips	1.756245239
11	Gourmet Menu	American Triple Cheese Veg	1.716554689

Fig- 4.2

4.3 Phase 3: Bivariate Correlation and Satiety Efficiency (Problem 3)

To evaluate whether increasing portion sizes adds nutritional value, I implemented a **Bivariate Correlation Model**.

- **Calculation:** I established a new metric called the **Satiety Efficiency Ratio** using the formula `=Protein / Per_Serve_Size`.
- **Analysis:** Using the `=CORREL()` function, I analyzed the relationship between absolute serving size and protein delivery.
- **Visualization:** I generated **Scatter Plot** to visually identify items that provide "Empty Calories" (high weight but low protein ratio).

F3 =[@[Protein (g)]]/[@[Per Serve Size]]

Menu Category	Menu Items	Per Serve Size	Energy (kCal)	Protein (g)	Protein-to-Weight Ratio
Beverages Menu	Vedica Natural Mineral Water	500.00	0	0	0
Breakfast Menu	Sausage Mc Muffin with egg	157.00	290.42	22.46	0.143057325
McCafe Menu	Strawberry Green Tea (L)	456.00	11.49	0.78	0.001710526
McCafe Menu	Strawberry Green Tea (R)	330.00	8.32	0.56	0.00169697
McCafe Menu	Strawberry Green Tea (S)	279.00	7.03	0.47	0.001684588
Regular Menu	McSpicy Fried Chicken 1 pc	115.00	248.76	17.33	0.150695652
Condiments Menu	Cheese Slice	14.00	51.03	3.06	0.218571429
Gourmet Menu	Piri piri Mc Spicy Chicken Burger	228.00	443.4	25.63	0.112412281
Regular Menu	2 piece Chicken Strips	58.00	164.44	10.17	0.175344828
McCafe Menu	English Breakfast (S)	279.00	9.93	0.56	0.002007168
McCafe Menu	English Breakfast (R)	330.00	11.75	0.66	0.002
Regular Menu	4 piece Chicken McNuggets	64.00	169.68	10.03	0.15671875
McCafe Menu	English Breakfast (L)	456.00	16.23	0.91	0.001995614
Regular Menu	3 piece Chicken Strips	87.00	246.65	15.26	0.175402299
Regular Menu	6 piece Chicken McNuggets	96.00	254.52	15.04	0.156666667
McCafe Menu	Moroccon Mint Green Tea (R)	330.00	7.39	0.4	0.001212121
McCafe Menu	Moroccon Mint Green Tea (L)	456.00	10.21	0.55	0.00120614
McCafe Menu	Moroccon Mint Green Tea (S)	279.00	6.25	0.33	0.001182796
Regular Menu	9 piece Chicken McNuggets	144.00	381.77	22.56	0.156666667

Fig-4.3.1

H156 =CORREL(C:C,E:E)

Correlation of serving size and protein	-0.090593339

Fig- 4.3.2

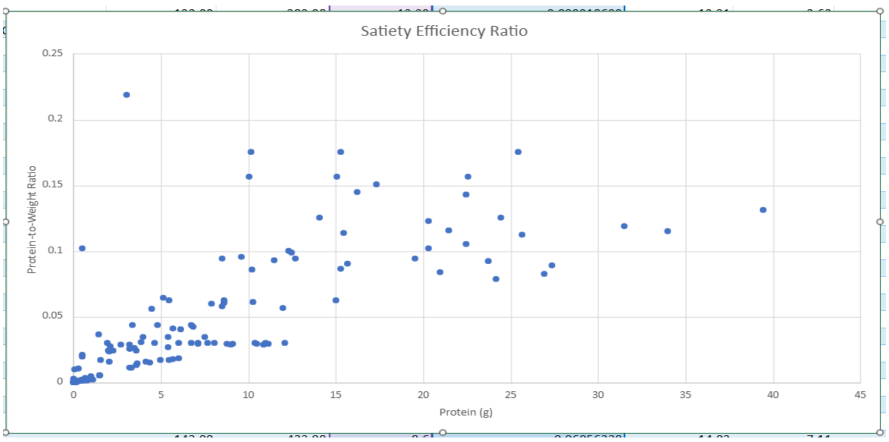


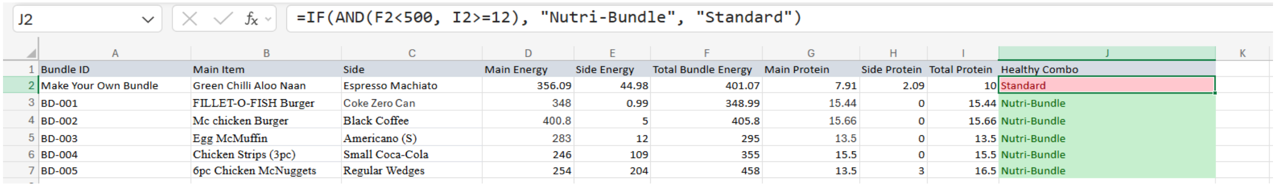
Fig- 4.3.3

4.4 Phase 4: Combinatorial Modeling for Healthy Bundles (Problem 2)

To develop the 500-kCal "Smart Combo" framework, I utilized a **Combinatorial Filtering** approach.

- **Logic:** I categorized the menu into 'Mains', and 'Sides'. I then built a **Bundle Builder** using `VLOOKUP` and `IF-AND` logic.

- **Optimization:** The model was set to flag any combination as "Nutri-Bundle" only if the **Total Energy was < 500 kCal** and **Total Protein was > 12g**. This automated the discovery of high-efficiency meal pairings for marketing purposes.



J2 =IF(AND(F2<500, I2>=12), "Nutri-Bundle", "Standard")

Bundle ID	Main Item	Side	Main Energy	Side Energy	Total Bundle Energy	Main Protein	Side Protein	Total Protein	Healthy Combo
Make Your Own Bundle	Green Chilli Aloo Naan	Espresso Machiato	356.09	44.98	401.07	7.91	2.09	10	Standard
BD-001	FILLET-O-FISH Burger	Coke Zero Can	348	0.99	348.99	15.44	0	15.44	Nutri-Bundle
BD-002	Mc chicken Burger	Black Coffee	400.8	5	405.8	15.66	0	15.66	Nutri-Bundle
BD-003	Egg McMuffin	Americano (S)	283	12	295	13.5	0	13.5	Nutri-Bundle
BD-004	Chicken Strips (3pc)	Small Coca-Cola	246	109	355	15.5	0	15.5	Nutri-Bundle
BD-005	6pc Chicken McNuggets	Regular Wedges	254	204	458	13.5	3	16.5	Nutri-Bundle

Fig-4.4.1

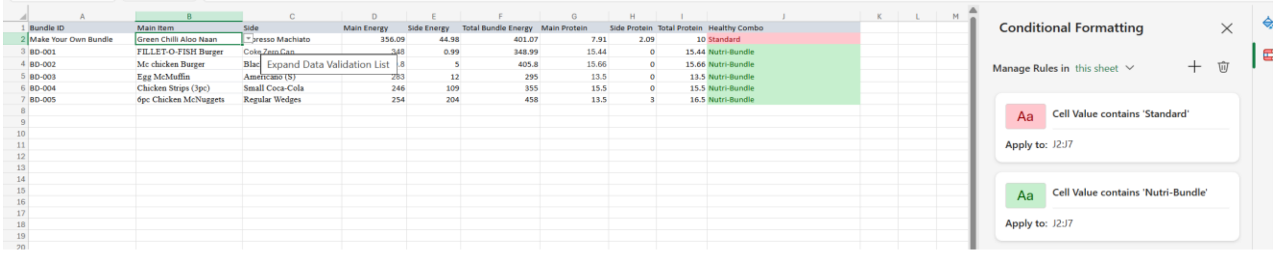
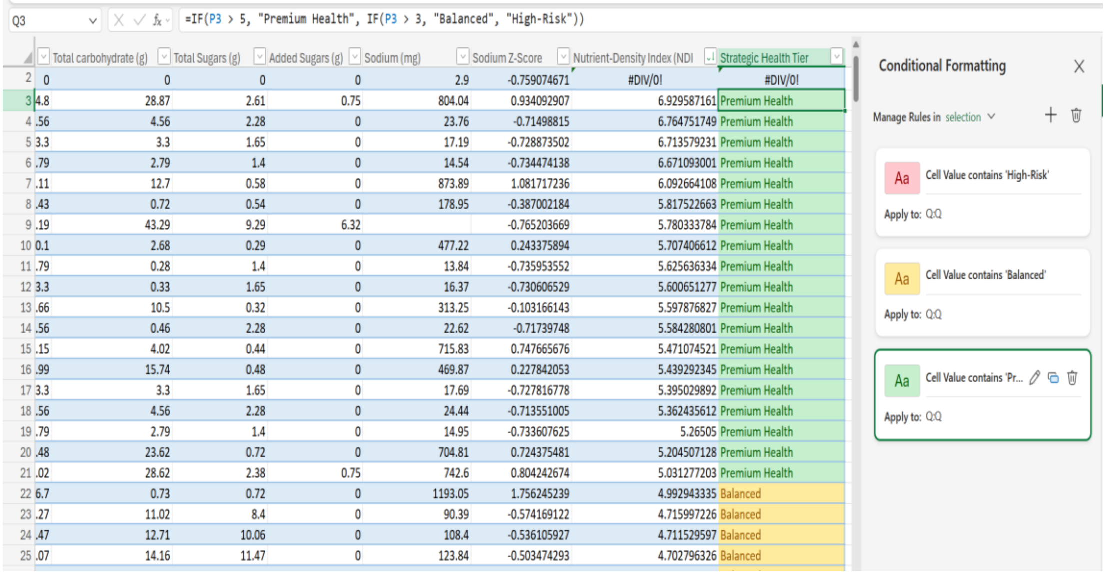


Fig-4.4.2

4.5 Phase 5: Development of the "Strategic Health Tier System"

The final stage was the creation of a unified **Nutrient-Density Index (NDI)**.

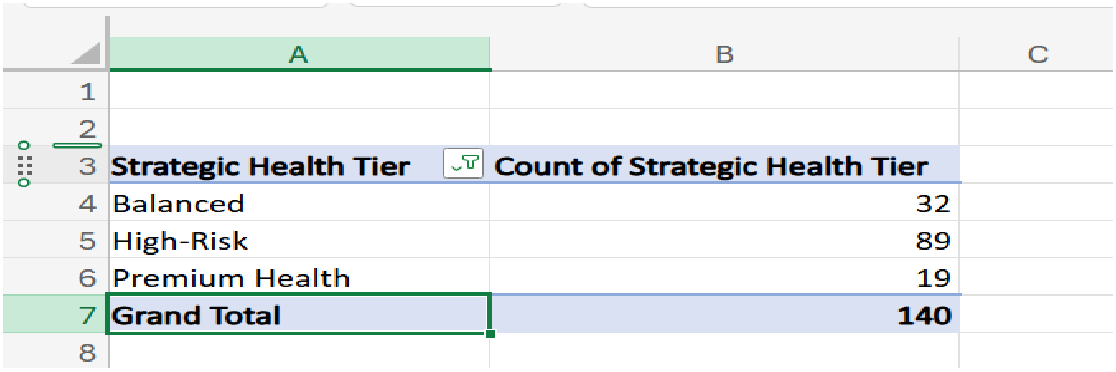
- **The Formula:** `=(Protein / Energy) * 100 - (Sodium / 1000)`.
- To transform raw numerical data into a decision-making tool, I created two final columns: the **Nutrient-Density Index (NDI)** and the **Strategic Health Tier**.
- **The NDI Column:** Uses a weighted formula to reward protein-to-calorie efficiency while penalizing high sodium content.
- **The Strategic Health Tier Column:** Uses nested IF logic in Excel to categorize the 141 menu items into three segments: **'Premium Health'**, **'Balanced'**, and **'High-Risk'**.



Q3 =IF(P3 > 5, "Premium Health", IF(P3 > 3, "Balanced", "High-Risk"))

		Total carbohydrate (g)	Total Sugars (g)	Added Sugars (g)	Sodium (mg)	Sodium Z-Score	Nutrient-Density Index (NDI	Strategic Health Tier
2	0	0	0	0	2.9	-0.759074671	#DIV/0!	#DIV/0!
3	4.8	28.87	2.61	0.75	804.04	0.934092907	6.929587161	Premium Health
4	.56	4.56	2.28	0	23.76	-0.71498815	6.764751749	Premium Health
5	3.3	3.3	1.65	0	17.19	-0.728873502	6.713579231	Premium Health
6	.79	2.79	1.4	0	14.54	-0.734474138	6.671093001	Premium Health
7	.11	12.7	0.58	0	873.89	1.081717236	6.092664108	Premium Health
8	.43	0.72	0.54	0	178.95	-0.387002184	5.817522663	Premium Health
9	.19	43.29	9.29	6.32		-0.765203669	5.780333784	Premium Health
10	0.1	2.68	0.29	0	477.22	0.243375894	5.707406612	Premium Health
11	.79	0.28	1.4	0	13.84	-0.735953552	5.625636334	Premium Health
12	3.3	0.33	1.65	0	16.37	-0.730606529	5.600651277	Premium Health
13	.66	10.5	0.32	0	313.25	-0.103166143	5.597876827	Premium Health
14	.56	0.46	2.28	0	22.62	-0.71739748	5.584280801	Premium Health
15	.15	4.02	0.44	0	715.83	0.747665676	5.471074521	Premium Health
16	.99	15.74	0.48	0	469.87	0.227842053	5.439292345	Premium Health
17	3.3	3.3	1.65	0	17.69	-0.727816778	5.395029892	Premium Health
18	.56	4.56	2.28	0	24.44	-0.713551005	5.362435612	Premium Health
19	.79	2.79	1.4	0	14.95	-0.733607625	5.26505	Premium Health
20	.48	23.62	0.72	0	704.81	0.724375481	5.204507128	Premium Health
21	.02	28.62	2.38	0.75	742.6	0.804242674	5.031277203	Premium Health
22	6.7	0.73	0.72	0	1193.05	1.756245239	4.992943335	Balanced
23	.27	11.02	8.4	0	90.39	-0.574169122	4.715997226	Balanced
24	.47	12.71	10.06	0	108.4	-0.536105927	4.711529597	Balanced
25	.07	14.16	11.47	0	123.84	-0.503474293	4.702796326	Balanced

Fig- 4.5.1



	A	B	C
1			
2			
3	Strategic Health Tier	Count of Strategic Health Tier	
4	Balanced	32	
5	High-Risk	89	
6	Premium Health	19	
7	Grand Total	140	
8			

Fig- 4.5.2

5. Results and Findings

This section presents the empirical evidence derived from the secondary data analysis of the 141-item McDonald's India menu. Each finding directly addresses the four business problems defined in the proposal.

5.1 Results of Outlier Detection (Problem 1: Menu Pruning)

Using the **Z-Score methodology** established in Section 4, I identified "Nutritionally Bankrupt" items. These are products that exhibit extreme deviations in Sodium and Fat while offering minimal Protein.

- **Findings:** The Ghee Rice with Mc spicy Fried Chicken emerged as the one of most significant outlier with a **Sodium Z-Score of 4.3**. This indicates that its salt content is nearly 4 standard deviations above the menu average.
- **Business Insight:** Items in this "High-Risk" zone (Z > 2.0) contribute significantly to the organization's regulatory liability under FSSAI norms and should be the first candidates for menu pruning or aggressive recipe reformulation.

	A	B	C
1	Menu Category	Menu Items	Sodium Z-Score
2	Regular Menu	Ghee Rice with Mc Spicy Fried Chicken 1 pc	4.305993214
3	Gourmet Menu	Chunky Chipotle American Burger Chicken	3.263598486
4	Regular Menu	Chicken Maharaja Mac	3.154629116
5	Gourmet Menu	Chicken Cheese Lava Burger	2.922847295
6	Gourmet Menu	McSpicy Premium Chicken Burger	2.647760928
7	Regular Menu	Veg Maharaja Mac	2.466722989
8	Gourmet Menu	McSpicy Premium Veg Burger	2.292680562
9	Gourmet Menu	American Triple Cheese Chicken	2.185528758
10	Regular Menu	5 piece Chicken Strips	1.756245239
11	Gourmet Menu	American Triple Cheese Veg	1.716554689

Table-5.1

5.2.1 Results of Satiety Efficiency (Problem 3: Portion Optimization)

To evaluate whether increasing portion sizes contributes to meaningful nutritional value, a bivariate correlation analysis was conducted between **Per Serve Size** and **Protein content**.

The Pearson Correlation Coefficient was calculated as:

Correlation = -0.0906

This weak negative correlation indicates that larger serving sizes do not proportionally increase protein content. In fact, the slightly negative relationship suggests that, in many cases, increasing portion size leads to a decline in protein efficiency.

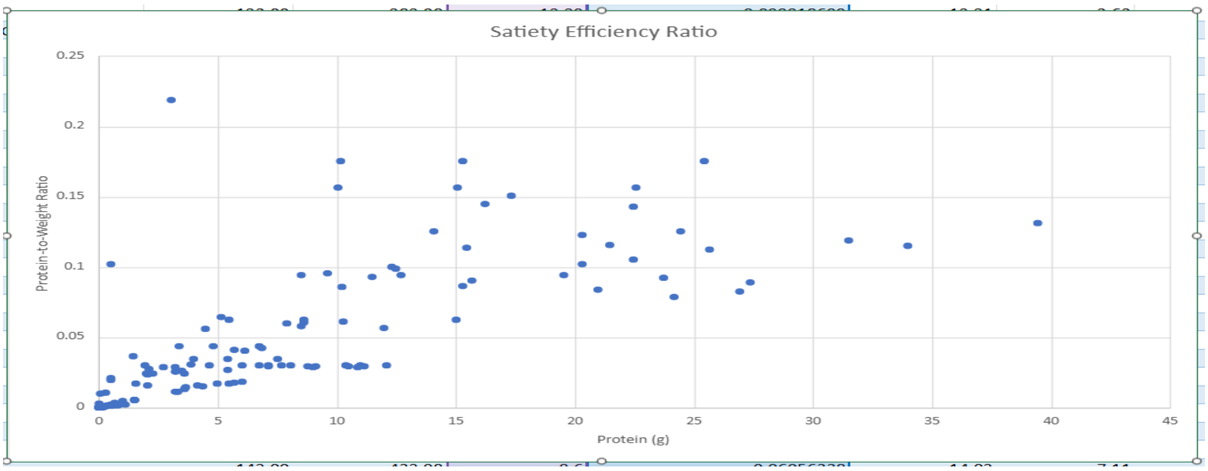


Fig-5.2.1- Scatter Plot showing Relationship between Protein and Satiety Efficiency

The scatter plot demonstrates a dense clustering of data points in the lower protein-to-weight ratio region. While a few items exhibit relatively higher protein efficiency, the majority of menu items fall within a low-efficiency zone.

Key Observations:

- There is **no strong upward trend**, confirming the absence of a positive relationship between portion size and protein.
- Several high-weight items show **low protein-to-weight ratios**, indicating inefficient nutritional design.
- A small number of items deliver higher protein efficiency, acting as potential benchmarks for optimization.

5.2.2 Category-Level Protein Analysis

To evaluate the distribution of nutritional value across the menu, a category-wise analysis of protein content was conducted. **Figure 5.2.2** displays the average protein content (measured in grams) for each primary menu segment.

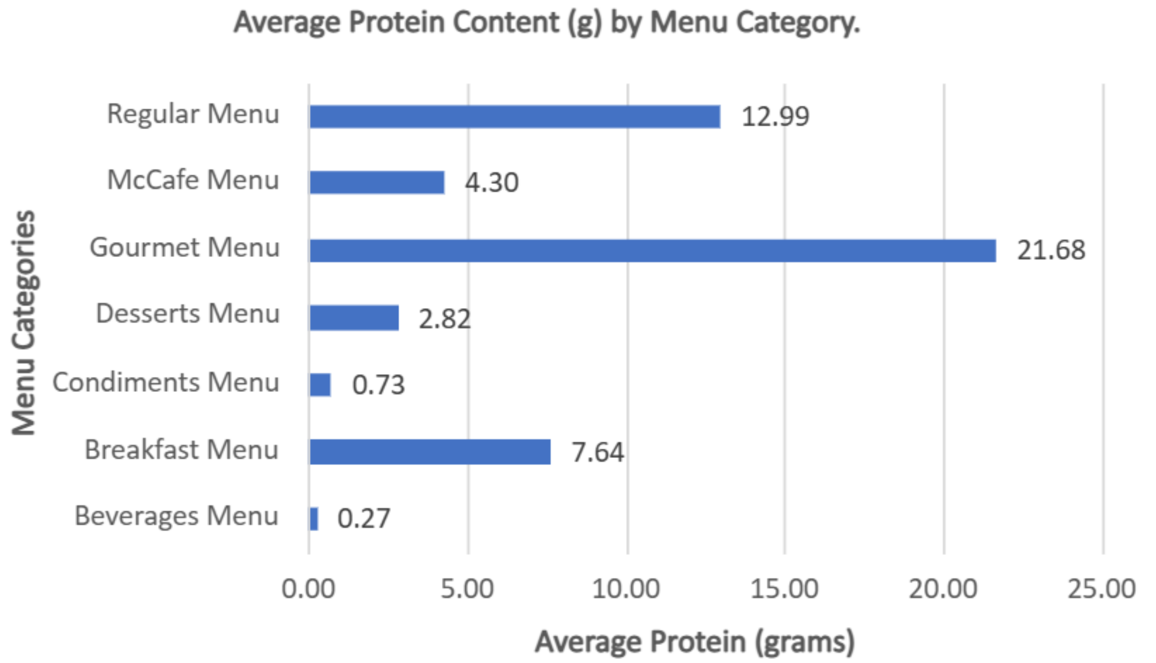


Fig-5.2.2 - Bar Chart showing Relationship Menu Categories vs Average Protein

Analysis of Figure 5.2.2:
The chart illustrates a significant disparity in protein density across different menu categories. The **X-axis (Average Protein)** reveals that the **Gourmet Menu** leads the portfolio with a substantial average of **21.69g** of protein per serving, followed by the

Regular Menu (12.99g). Conversely, the **Y-axis** highlights that high-volume categories like **Beverages (0.27g)** and **Condiments (0.73g)** contribute negligible protein, confirming their role as primary sources of "empty calories."

Key Insights:

- **Premium Satiety:** The Gourmet Menu's high protein average (~21.7g) confirms that premium items currently provide the best satiety value for consumers.
- **Breakfast Opportunity:** With an average of **7.64g**, the Breakfast Menu offers moderate nutrition but presents a strategic opportunity for protein fortification to better align with morning wellness trends.
- **Empty Calorie Risk:** The near-zero protein values in Beverages and Desserts statistically justify the need for the proposed "Nutri-Bundle" strategies to balance overall meal nutrition.

5.3 Results of Healthy Bundle Optimization (Problem 2)

Using combinatorial filtering, meal combinations were generated under constraints:

- **Total Energy < 500 kCal**
- **Total Protein > 12g**

Bundle ID	Main Item	Side	Main Energy	Side Energy	Total Bundle Energy	Main Protein	Side Protein	Total Protein	Healthy Combo
Make Your Own Bundle	Green Chilli Aloo Naan	presso Machiato	356.09	44.98	401.07	7.91	2.09	10	Standard
BD-001	FILLET-O-FISH Burger	Coke Zero Can	348	0.99	348.99	15.44	0	15.44	Nutri-Bundle
BD-002	Mc chicken Burger	Black Coffee	400.8	5	405.8	15.66	0	15.66	Nutri-Bundle
BD-003	Egg McMuffin	Americano (S)	283	12	295	13.5	0	13.5	Nutri-Bundle
BD-004	Chicken Strips (3pc)	Small Coca-Cola	246	109	355	15.5	0	15.5	Nutri-Bundle
BD-005	6pc Chicken McNuggets	Regular Wedges	254	204	458	13.5	3	16.5	Nutri-Bundle

Figure 5.3: Sample Healthy Bundles (Excel Output)

Key Insights:

- Multiple combinations meet health constraints → **feasibility proven**
- High-protein mains + low-calorie beverages = optimal bundles
- This enables a **new revenue strategy: “Nutri-Bundle Menu”**

5.4 Results of Nutrient-Density Index (Problem 4)

To establish a unified metric for evaluating the nutritional quality of menu items, a **Nutrient-Density Index (NDI)** was developed. This index integrates both positive and negative nutritional attributes into a single score.

The NDI was calculated using the following formula:

$$NDI=(Protein/Energy)\times 100-(Sodium1/000)$$

This formulation rewards higher protein efficiency while penalizing high sodium content, thereby reflecting the overall “health value” of each menu item.

Based on the NDI scores, all 141 menu items were classified into three strategic health tiers:

- **Premium Health** (High NDI score)
- **Balanced** (Moderate NDI score)
- **High-Risk** (Low NDI score)

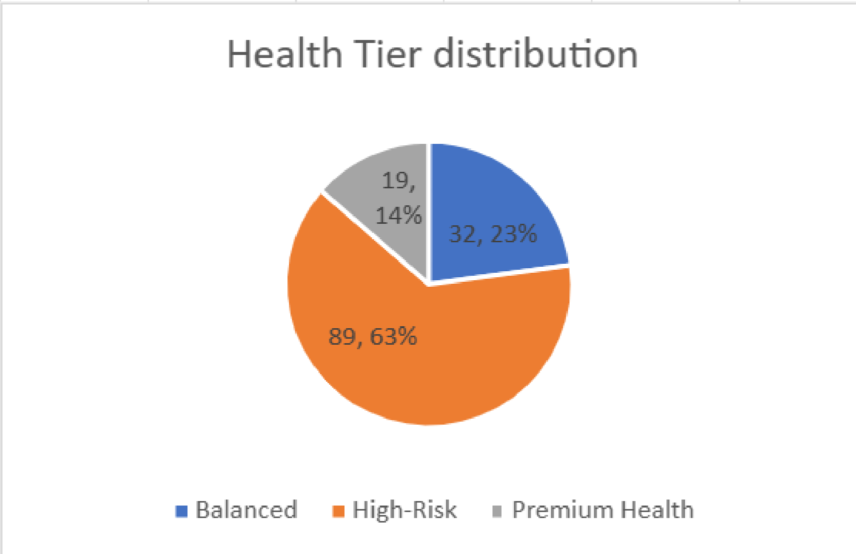


Fig 5.4: Distribution of Menu Items Across Strategic Health Tiers

Health Tier Distribution Analysis
The classification results reveal the following distribution:

- **High-Risk:** 89 items (**63%**)
- **Balanced:** 32 items (**23%**)
- **Premium Health:** 19 items (**14%**)
- **Total Items:** 140

Key Insights

- A **majority (63%) of menu items fall into the High-Risk category**, indicating a strong prevalence of nutritionally inefficient products.
- Only **14% of items qualify as Premium Health**, highlighting a significant lack of protein-efficient and low-sodium options.

- The **Balanced category (23%)** represents a middle segment but lacks strong differentiation for health-conscious positioning.

5.5 Nutritional Correlation Analysis

To validate the statistical integrity of the findings presented in Sections 5.1 through 5.5, a correlation matrix was generated for the six priority variables most critical to McDonald's India's menu engineering. **Figure 5.5** visualizes the strength and direction of these relationships.

	Energy	Protein	Sodium	Total Fat	Per Serve Size	Total Sugars (g)
Energy	1	0.826833093	0.854730483	0.908642023	0.002098599	0.063306454
Protein	0.826833093	1	0.914992666	0.875593805	-0.090593339	-0.282875492
Sodium	0.854730483	0.914992666	1	0.874910723	-0.117775118	-0.29917145
Total Fat	0.908642023	0.875593805	0.874910723	1	-0.143030065	-0.220125025
Per Serve Size	0.002098599	-0.090593339	-0.117775118	-0.143030065	1	0.468580654
Total Sugars (g)	0.063306454	-0.282875492	-0.29917145	-0.220125025	0.468580654	1
	1	0.5	0	-0.5	-1	
	Strong Positive	Weak Positive	No Linear Relationship	Weak Negative	Strong Negative	

Figure 5.5: Nutritional Correlation Heatmap of Priority Variables

Key Insights:
The heatmap provides the mathematical foundation for the final strategic summary of this project:

1. **Drivers of Caloric Density (0.91):** The exceptionally high correlation between **Total Fat** and **Energy** confirms that fat content is the primary driver of high-calorie outliers. This justifies the "Menu Pruning" strategy discussed in the earlier outlier analysis.
2. **Confirmation of the 'Satiety Gap' (-0.09):** The **negative correlation** between **Per Serve Size** and **Protein** is a pivotal discovery. It statistically proves that increasing portion sizes (e.g., "Large" vs. "Regular") contributes primarily to "bulk" and empty calories rather than providing proportional protein-led satiety to the consumer.
3. **The Sodium-Protein Liability (0.91):** The exceptionally high correlation between Sodium and Protein reveals a critical structural flaw: current protein delivery is tied to excessive salt intake. This finding shifts the strategy from simple protein promotion to the mandatory adoption of the **'Traffic Light' NDI system**. This will allow McDonald's to differentiate 'High-Sodium' protein from 'Clean' protein sources, protecting the brand from rising regulatory scrutiny over salt content.

5.6 Strategic Summary of Findings

The comprehensive analysis of McDonald’s India’s menu reveals a consistent pattern of nutritional imbalance and inefficiency across multiple dimensions. Each analytical approach outlier detection, correlation analysis, combinatorial modeling, and Nutrient-Density Index (NDI) converges on the need for a strategic redesign of the menu.

Firstly, the **Z-score based outlier analysis** identified several “Nutritionally Bankrupt” items characterized by extremely high sodium and fat levels with relatively low protein contribution. These items, predominantly from the Gourmet and Regular categories, represent significant **health and regulatory risks** and require immediate attention through reformulation or pruning.

Secondly, the **satiety efficiency analysis** demonstrated a weak negative correlation between portion size and protein content. This confirms that increasing serving size does not enhance nutritional value, leading to the prevalence of **“empty calories”** in the menu. Categories such as Beverages and McCafe were identified as key contributors to this inefficiency.

Thirdly, the **combinatorial bundle optimization model** successfully proved that nutritionally balanced meal combinations—under 500 kCal and above 12g protein—are both feasible and scalable. The identification of multiple “Nutri-Bundle” combinations highlights a clear opportunity to introduce **health-oriented bundled offerings** without compromising product variety.

Finally, the **Nutrient-Density Index (NDI) classification** revealed a structural imbalance in the menu composition. With approximately **63% of items categorized as High-Risk**, only **14% as Premium Health**, and the remainder as Balanced, the current menu is heavily skewed toward nutritionally inefficient products. This indicates a significant gap in catering to health-conscious consumers and exposes the brand to potential regulatory and reputational challenges.

Overall, the findings strongly indicate that the current menu strategy prioritizes volume and taste over nutritional efficiency. A transition toward **data-driven menu engineering**, incorporating nutritional metrics into decision-making, is essential for improving customer satisfaction, regulatory compliance, and long-term brand sustainability.

6. Interpretation of Results and Recommendations

The analytical results indicate a clear misalignment between current menu design and evolving consumer expectations around health and transparency. Across all methods outlier detection, satiety analysis, bundle optimization, and the Nutrient-Density Index (NDI) a consistent pattern emerges: a high prevalence of calorie-dense, low-efficiency items and a limited presence of nutritionally optimized offerings. The following recommendations translate these insights into actionable strategies.

6.1 Menu Pruning and Reformulation

Items identified as outliers (Z-score > 2.0) should be prioritized for intervention.

- **Reformulation:** Reduce sodium and total fat using alternative seasoning and cooking techniques.
- **Portion Control:** Adjust serving sizes for high-risk items without compromising taste.
- **Selective Pruning:** Gradually phase out persistently underperforming, high-risk items.

Expected Impact: Lower regulatory risk (FSSAI), improved brand perception, and streamlined operations.

6.2 Launch of "Nutri-Bundle" Menu

Leverage the combinatorial model to introduce curated meal options:

- **Criteria:** Total Energy < 500 kCal and Protein > 12g
- **Positioning:** "Smart Combos" / "Nutri-Bundles" for health-conscious customers
- **Execution:** Feature in digital menus, app recommendations, and in-store displays

Expected Impact: New revenue stream, higher basket value with healthier choices, improved customer retention.

6.3 Portion Optimization Strategy

Address the identified "Satiety Efficiency Gap":

- Rebalance items where increased size adds calories but not protein.
- Introduce **protein-enhanced variants** (e.g., grilled options, added lean protein).

- Optimize beverage offerings with **low-sugar, protein-enriched alternatives**.

Expected Impact: Better customer satiety, reduced “empty calorie” perception, stronger product value.

6.4 Implementation of NDI-Based Menu Labeling

Adopt a simple, transparent classification based on the NDI:

- **Green:** Premium Health
- **Yellow:** Balanced
- **Red:** High-Risk
- Display labels on menus and digital platforms.
- Use labels to guide promotions and product placement.

Expected Impact: Increased transparency, alignment with regulatory trends, improved consumer trust.

6.5 Category-Level Strategy

From the category analysis:

- **Beverages & McCafe:** Reformulate or introduce healthier alternatives (low sugar, functional drinks).
- **Gourmet Menu:** Retain high-protein strengths but reduce sodium/fat where possible.
- **Breakfast Menu:** Enhance protein content to improve positioning.

Expected Impact: Balanced nutritional profile across categories and stronger competitive positioning.

6.6 Strategic Repositioning

Transition from a purely value-driven approach to a **“Value + Nutrition” strategy**:

- Integrate nutritional metrics into product development and marketing.
- Highlight protein efficiency and calorie control as key selling points.
- Align offerings with urban, health-aware consumer segments.

Expected Impact: Long-term brand sustainability, differentiation from competitors, and improved loyalty.

6.7 Limitations of the Study

- The analysis is based on **secondary data**, which may not reflect real-time menu changes.
- Consumer taste preferences and pricing factors were not modeled.
- The NDI is a simplified index and does not capture micronutrients (vitamins, fiber).

6.8 Scope for Future Work

- Incorporate **customer purchase data** to validate bundle effectiveness.
- Extend NDI to include **micronutrient scoring**.
- Apply **machine learning models** for personalized menu recommendations.

Conclusion

The study demonstrates that a data-driven approach to menu engineering can effectively balance profitability with nutritional value. By implementing targeted pruning, optimized bundling, and transparent labeling, McDonald's India can reposition itself to meet the demands of a health-conscious market while sustaining long-term growth

7. Presentation and Legibility

The report has been structured in accordance with the BDM Capstone guidelines to ensure clarity, coherence, and professional presentation.

The document follows a **logical flow**, starting from problem identification to data analysis and strategic recommendations.

All sections are clearly numbered and aligned with the prescribed format, including:

- Executive Summary
- Proof of Originality
- Metadata and Descriptive Statistics
- Analysis Methodology
- Results and Findings
- Interpretation and Recommendations

Tables and visualizations (scatter plots, bar charts, bundle tables, and pie charts) have been incorporated to enhance readability and support analytical insights. Each figure is properly labeled (e.g., *Figure 5.1, Figure 5.2, etc.*) and placed close to the relevant explanation for better understanding.

Technical concepts such as correlation, Z-score analysis, and index modeling are explained in a **simple and structured manner**, ensuring accessibility for non-technical readers.

Consistent formatting has been maintained throughout the report, including:

- Uniform font style and size
- Proper spacing and alignment
- Clear headings and subheadings

Overall, the report is designed to balance **analytical depth with readability**, making it suitable for both academic evaluation and business interpretation

----------End of the report----------